SITUATIONAL AI

An Emerging Programming Paradigm

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Research Purpose & Central Question

Research Purpose

To determine whether Situational AI represents an emerging programming paradigm in which situations, not prompts, tasks, or workflows, serve as the core abstraction for AI reasoning and execution.

Research Question

Does Situational AI represent an emerging programming paradigm in which situations, rather than prompts, tasks, or workflows, serve as the core abstraction for reasoning, decision-making, and execution?

Why It Matters

- Current AI systems process prompts and execute workflows but lack persistent situational understanding.
- This paper names the missing architectural layer and proposes situations as the primary unit of AI intelligence.
- If valid, this shift marks a new stage in software design beyond agentic orchestration.

Sources: Van Roy (2009); Sawant et al. (2021); Yu et al. (2025); A Survey on Agent Workflow (2025)

Background: What Is Situational AI?

A programming paradigm in which a situation, not a prompt, task, or workflow, becomes the primary unit of AI reasoning, decision-making, and execution. The system interprets context, state, priorities, expected response, and escalation conditions as one connected frame.



Traditional Software

Fixed rules, procedural logic, OOP. Behavior is decided in advance by the developer. No runtime adaptation.



ML-Enabled Systems

Patterns learned from data. Adapts via training pipelines, monitoring, and lifecycle management.



Situational AI

Situations as the primary abstraction. Context, state, escalation, and goals maintained as one persistent frame.

Sources: Van Roy (2009); O'Connor et al.; Buyya et al.; Sawant et al. (2021); Zhang et al. (2019); Sanneman et al. (2020)

Problem Significance: The Gap in AI Design

Why This Problem Is Relevant

Episodic AI is insufficient

Prompt-based systems reset with each interaction, no continuity.

Orchestration ≠ Understanding

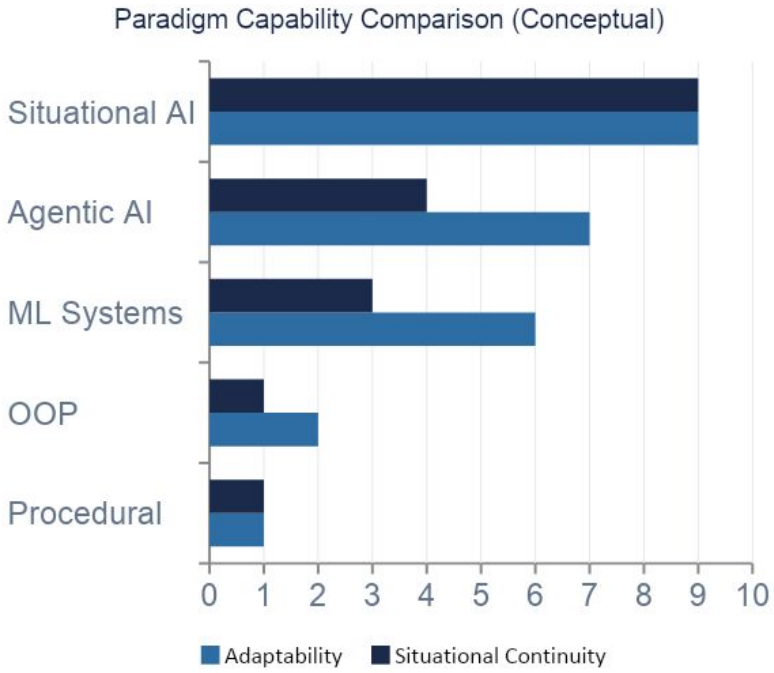
Agent workflows manage steps, but do not reason from the situation.

ML pipelines need humans

Even advanced systems rely on human-maintained lifecycles (Nahar et al., 2022).

Reasoning breaks under complexity

LLM reasoning quality collapses as task complexity grows (Apple, 2026).



Sources: Nahar et al. (2022); Apple Machine Learning Research (2026); Yu et al. (2025); Adimulam et al. (2026)

Review Depth & Breadth

Traditional Paradigms

Procedural, OOP, fixed-rule systems; baseline for comparison.

ML-Enabled Systems

Data pipelines, lifecycle management, deployment, monitoring.

Agentic & AI-Native

Orchestration, tool use, workflow decomp, governance, trust.

Situational Awareness

Perception, comprehension, projection, intellectual precursors.

Situational AI Architecture

Situation memory, escalation logic, response planning, explainability.

Human-AI Coworker Systems

Bounded autonomy, monitoring, prioritization, human oversight.

Review spans: Van Roy (2009) through Yu et al. (2025), Adimulam et al. (2026), Sanneman et al. (2020), Kaufman et al. (2024), and 30+ additional sources.

Results: The Paradigm Progression

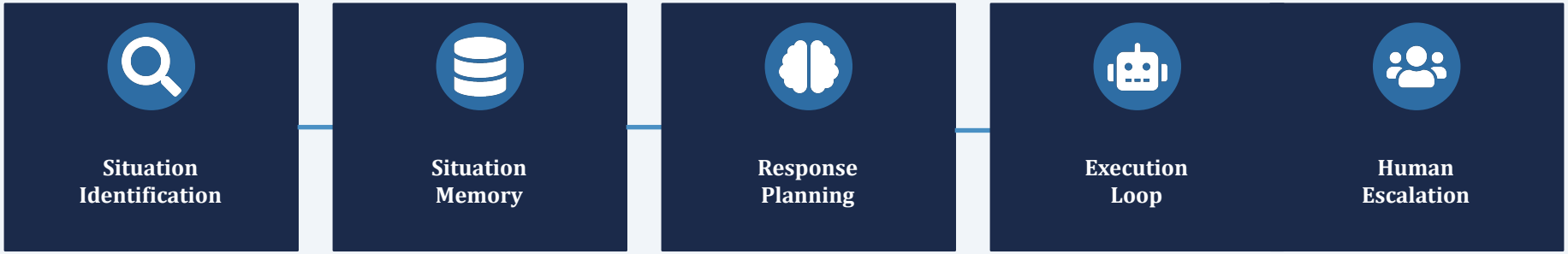
Procedural & OOP	ML Systems	Agentic AI	Situational AI
Pre-2000s	*2000s–2015*	*2015–2024*	*Emerging*
Fixed control flow; behavior fully specified in advance	Infers patterns from data; adapts via training & monitoring	Orchestrates tools, tasks, and multi-step workflows	Situation as primary abstraction; persistent context & reasoning

Key Findings

1. Traditional and ML paradigms lack a persistent situational layer, behavior is either predetermined or pipeline-managed.
2. Agentic systems improve capability through orchestration but remain workflow-based rather than situation-centered (Yu et al., 2025; Adimulam et al., 2026).
3. Situational AI introduces situation as the primary unit: context, state, priorities, escalation, and response logic held together as one frame.

Sources: Van Roy (2009); Sawant et al. (2021); Yu et al. (2025); Adimulam et al. (2026); Apple ML Research (2026); Yufik & Malhotra (2021)

Concrete System Model: Situational AI Architecture



Explainability

System must make reasoning legible for human review (Sanneman et al., 2020; Bansal et al., 2023).

Layered Cognition

Perception, symbolic structure, and reasoning kept distinct - not compressed into a single process.

Bounded Autonomy

Autonomy varies with the situation; escalation to human oversight is a first-class design requirement.

Sources: Sanneman et al. (2020); Bansal et al. (2023); Zhang et al. (2019); A Conceptual Framework for AI-based Decision Systems (2025)

Potential Research Methodology (Future Empirical Work)

01

Prototype Implementation

Build a Situational AI prototype incorporating situation identification, memory, response planning, execution, and escalation. Compare against a baseline agentic system on matched tasks.

02

Controlled Task Study

Deploy both systems on tasks of increasing complexity and novelty. Measure: situation recognition accuracy, context retention, escalation precision, and task success rate.

03

Human-AI Teaming Evaluation

Embed the system in a coworker context. Evaluate trust, explainability, workload distribution, and operator cognitive load using validated scales (NASA-TLX, SART).

04

Paradigm Validation

Conduct expert panel reviews with AI researchers and software architects. Assess whether Situational AI meets paradigm criteria: distinct abstraction, generativity, and adoption potential.

Proposed methodology informed by: Nahar et al. (2022); Sanneman et al. (2020); Bansal et al. (2023); Hagemann et al. (2023)

Limitations of the Review

Limitation	Description	Severity
No empirical validation	Situational AI remains a conceptual framework. No prototype has been built or tested against existing agentic systems.	**High**
Paradigm overlap	Existing cognitive and agentic architectures already cover parts of the proposed space (Hagemann et al., 2023).	**Medium**
Emerging literature	Core sources (2024–2026) are recent. Peer review and replication are still limited across the domain.	**Medium**
Definitional ambiguity	"Situation" lacks a formal computational definition. Boundary conditions vs. prompts and tasks are not yet precisely specified.	**High**
Safety & oversight unresolved	Meaningful governance requires bounded autonomy in real time (Manheim & Homewood, 2025), not yet addressed architecturally.	**High**

Sources: Hagemann et al. (2023); Manheim & Homewood (2025); Wen et al. (2023); Nofal et al. (2023)

Counterarguments & Scholarly Balance

Supporting the Paradigm Claim

✓ Situational awareness research already identifies responsiveness and context as central (Hagemann et al., 2023).

✓ Human-AI teaming requires shared situational models beyond what workflows provide (Wen et al., 2023).

✓ Current architectures are explicitly described as orchestration-heavy, not situation-centered (Yu et al.; Adimulam et al.).

✓ The literature consistently points to a missing persistent context layer as the unsolved problem.

Challenges to the Paradigm Claim

✗ May be an extension of existing human-AI teaming research rather than a wholly new paradigm (Hagemann et al., 2023).

✗ Cognitive interaction layers and neuro-symbolic AI already cover portions of the proposed space.

✗ No formal computational definition of "situation" yet, boundary conditions are unresolved.

✗ Oversight and control remain open questions; bounded autonomy in real time is not yet architecturally solved (Manheim & Homewood, 2025).

Sources: Hagemann et al. (2023); Wen et al. (2023); Nofal et al. (2023); Manheim & Homewood (2025)

Future Research Directions

Formal Computational Definition

Develop a precise, testable definition of 'situation' as a computational unit, distinguishing it from prompts, tasks, context windows, and workflows.

Prototype & Empirical Benchmarking

Build and evaluate a Situational AI prototype against agentic baselines on tasks requiring continuity, novelty, and escalation.

Human-AI Coworker Deployment

Test Situational AI in real human-AI teaming environments, measuring trust, explainability, cognitive load, and decision quality.

Governance & Safety Frameworks

Address bounded autonomy, oversight, and control mechanisms that allow Situational AI to operate safely in high-stakes settings (Manheim & Homewood, 2025).

Paradigm Adoption Conditions

Investigate whether Situational AI meets formal criteria for paradigm adoption: distinct abstraction, generativity, tooling, and community formation.

Cross-Domain Application

Explore Situational AI in healthcare, critical infrastructure, and autonomous systems where persistent situational understanding is most consequential.

Sources: Manheim & Homewood (2025); Sanneman et al. (2020); Hagemann et al. (2023); Human-centric AI Architecture (2023)

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Thank you